Music Therapy Brain Computer System (MT-BCI) Information Sheet

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Dear Participant,

Thank you for your support to our MT-BCI study. This study is being conducted at Otto von Guericke University Magdeburg under the supervision of Prof. Sebastian Stober. This document gives information about the study and experiment process. Participation in these experiments is on a voluntary basis. Before you decide to participate, please read the document carefully and do not hesitate to ask question if you need.

1 Purpose of the Study:

This study aims to analyze participants' mental state and accordingly recommend music in order to regulate their mental states. The entire system is called as affective Brain Computer Interface (aBCI) and it measures brain signals using electroencephalogram (EEG) technology, analyze mental state from EEG signal and recommend music. The detailed experiment procedure is described in the following sections.

2 EEG Gathering Process:

EEG is one of the most used non-invasive tools in cognitive science researches to record brain activities. It measures the electrical activities of the brain by placing electrodes on the scalp surface. In order to get good connectivity, the gel is applied between electrode and scalp. Therefore, it is important to come experiments washed and dried hair without any conditioner. Additionally, you will need to wash your hair after experiments.

3 Self Assessment Manikin (SAM):

All emotions can be described as a linear combination of two underlying independent neurophysiological systems; valance and arousal (Russell, 1980). Based on this idea, self assessment manikin is proposed (Bradley and Lang, 1994). Arousal score represents the emotion from calm and excited, and valance score represents the emotion from unpleasant to pleasant.

In the Figure 1 SAM scale figures are represented for each dimension. For valance dimension, negative/unpleasant score is 1 and positive/pleasant score is 9. Similarly for arousal dimension, calm/relaxed score is 1 and excited/tension score is 9.

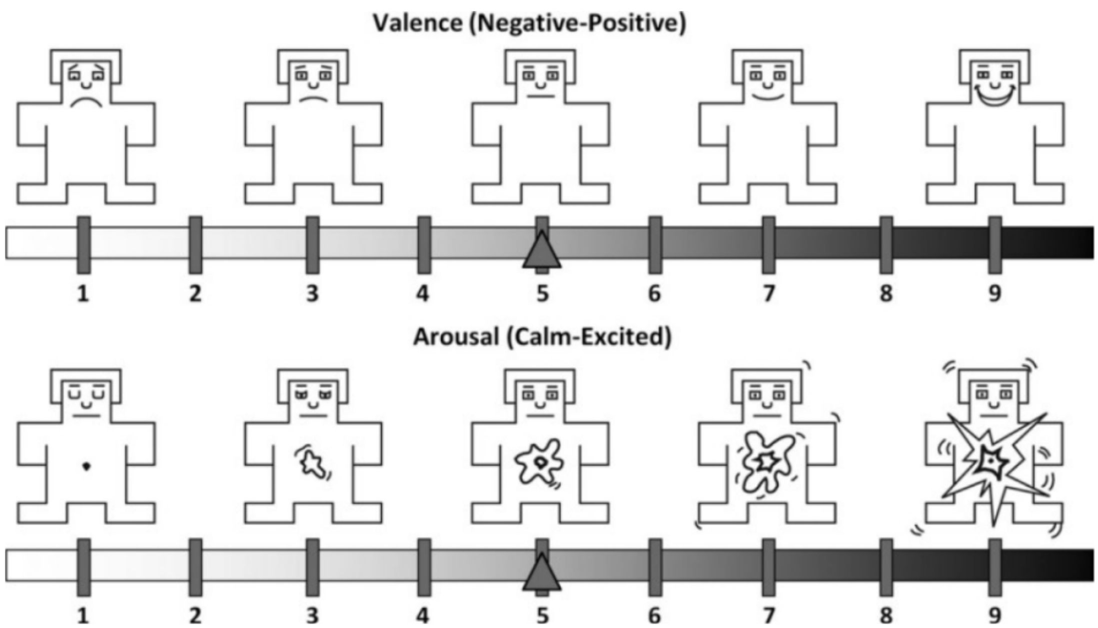

Figure 1: Self Assessment Manikin

4 Experiment Process:

You can find the block diagram of experiment process in the Figure 2.

- Please read the document before you come to the experiments and ask questions if necessary.
- [$5min$] Please sign the consent form and complete the questionnaire
- [$25min$] The EEG cap will put on and gel will be applied. This process may take some time to be sure the consistency of the system.
- During gel application, you are shown two different song lists from sad and happy categories. Please select 3 songs from each category. You can listen songs as you want and select that makes you sad and happy.
- [$2min$] You will practice music listening and assessing your emotion for a short time to get used to the system.
- For every selected song we will do following steps; (in total 6 trials)
  - [$10sec$] Assess your emotion before listening to music using SAM methodology described above. Please select a score between 1 and 9 for each valance and arousal dimensions
  - [$20sec$] Listening to one of the selected music. Please look to the cross sign on the screen during music play
  - [$10sec$] Assess your emotion after listening to music using SAM methodology. Please select a score between 1 and 9 for each valance and arousal dimensions
  - [$10sec$] Before we start new trial, there will be an idle session. Here there is no music and fixation cross. There will be a neutral image on the screen.
- The above-mentioned steps for all trials constitute one run, and this experiment consist of 5 runs.
- The overall experiment process will take 57 minutes.

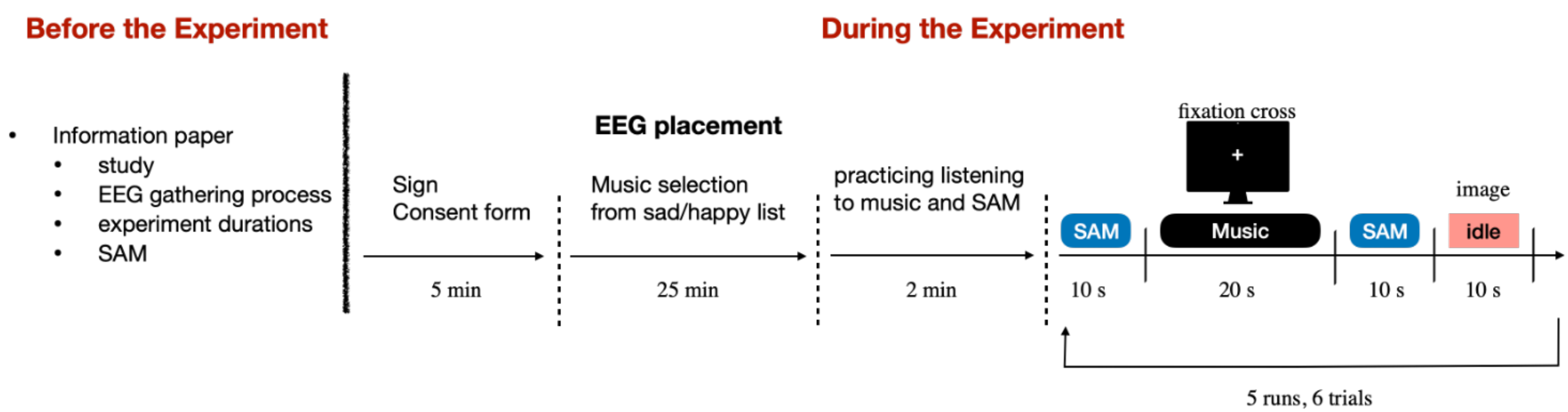

Figure 2: The block diagram of calibration experiment design